

Project: **Underwater search and navigation in underwater realistic environment.**

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The preservation of marine environments is a strategic priority for coastal regions, especially in light of the increasing human impact from industrialization, agriculture, offshore oil and gas activities, and desalination processes. The Arabian Gulf, one of the most saline and warm marine environments in the world, is under growing pressure. Rising sea surface temperatures, Coral bleaching and mortality, and human activities are key concerns. Continuous, precise, and non-invasive monitoring is essential for assessing ecosystem health. Swarms of underwater robots are particularly effective for covering large areas. **This master's thesis aims to develop adaptive control algorithms (specifically multi-agents reinforcement learning) to enhance the autonomy of multi-robot systems for improved monitoring of marine flora and fauna.** The control algorithms will be tested in Abu Dhabi coastal waters.

During the planning phase of artificial coral reef restoration, there is a need to identify the right locations for **planting the new coral reef based on the presence of human activities and the most ideal physical parameters.** The criteria are based on keeping a safe distance from areas actively used for industrial or commercial purposes (industrial discharges, desalination plants, harbors, oil&gas facilities) and measuring temperature, turbidity, and light levels.

To overcome the costly and time consuming methods based on diver missions, a **key focus of the project is the development of collaborative algorithms that will enable the AUVs to operate as a coordinated autonomous mobile underwater sensor network able to collect data of wide areas in a short time.** Each AUV can measure physical, chemical and biological parameters. Exploiting the numerosity of the AUVs, the mission time can be reduced. **Since the mission duration is extremely relevant mostly in the underwater environment, being constrained by the battery life and impacting on the mission costs, we want to improve the performance of the collective mission, leveraging on information sharing, collective intelligence and optimized strategies oriented to identify the best location in the shortest possible time.** Multi-agents reinforcement learning (MARL) appear to be the ideal solution for this kind of task, characterized by uncertainty of the environment, and need of adaptability to changing environmental conditions. Applying MARL to this problem represents a novel contribution to the state of the art.

The project will be developed in collaboration with Sorbonne University Abu Dhabi (UAE). Sorbonne University Abu Dhabi (SUAD) has declared 2025 as the Year of the Ocean, reaffirming its commitment to marine conservation, research, and sustainability. This initiative aligns with the Third UN Ocean Conference (UNOC3), which will take place in Nice, France, from 9-13 June 2025. Sorbonne University also hosts the Ocean Institute, a multidisciplinary center focused on ocean research.

The Master thesis will be developed in University of Trieste under the supervision of Prof. Antonio Celani and co-supervision of Dr. Giulia De Masi.

This Master thesis is in part of the larger IREG Project "Collective Artificial Intelligence for Multi-Robot Coral Reef Monitoring and Habitat Enhancement: A comparison across Gulf countries", funded by Sorbonne University.

A previous thesis explored already the algorithmic aspect of MARL to achieve efficient search of physical quantities.

In parallel the robotic turtles are under design, development and realization in Abu Dhabi being very close to a first working prototype.

The aim of the current thesis project is to include the physics of the underwater environment that can be of high impact on the navigation and very relevant also in terms of realism for robotics system functioning and energy consumption.

The project will include:

- preliminary assessment of existing simulators: In particular MarineGym and SwarmSim
- Implementation of two/three robotic agents for searching scenario like described above
- Consideration on energy efficiency

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